



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 • STANDARD 6.DS.4

## Describe Data Using the Mean

Lesson 2-8 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

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## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can determine the mean of a data set, use a target mean to find a missing value, and explain the mean as the fair share and the balance point of the data.

**Language Objective:** I can explain my reasoning using the words mean, average, measure of center, balance point, and data set.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Mean

Average

Measure of center

Balance point

Data set



Tap any word to see what it means and a picture.

1

The value found by adding all the data and dividing by how many values there are is the .

2

A common everyday word that means the same thing as the mean is the .

3

A single number that describes the typical value of a whole data set is a .

4

The value where the total amount above equals the total amount below is the .

5

The whole collection of collected values is the .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

COMPARE

**How many points does Sofia need to score in the last game to hit her goal, and how does the balance-point view give the same answer as adding and dividing?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Mean** — The average of a data set. Add all the values, then divide by how many values there are.  
*Media* — El promedio de un conjunto de datos. Suma todos los valores y divide entre cuántos valores hay.

- ☐ **Average** — Another word for the mean — the amount each value would be if the total were shared equally.  
*Promedio* — Otra palabra para la media: la cantidad que tendría cada valor si el total se repartiera en partes iguales.

- ☐ **Measure of center** — A single number used to describe the middle, or typical value, of a whole data set.  
*Medida de centro* — Un solo número que se usa para describir el centro, o valor típico, de todo un conjunto de datos.

- ☐ **Balance point** — The value where the total distance of the data above it equals the total distance below it. The mean is the balance point.  
*Punto de equilibrio* — El valor donde la distancia total de los datos por encima es igual a la distancia total por debajo. La media es el punto de equilibrio.

- ☐ **Data set** — The whole collection of values that were gathered or measured.  
*Conjunto de datos* — Toda la colección de valores que se recopilaron o midieron.

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**After 9 games Sofia has scored \_\_\_\_ points. To average 15 across 10 games she needs \_\_\_\_ points in all, so game 10 must be \_\_\_\_ points. Using the balance point instead, her games total \_\_\_\_ points below the target mean and \_\_\_\_ points above it, so game 10 has to land \_\_\_\_ the target mean.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** A typical day is the amount Miguel would earn if he earned the SAME amount every day.

**Evidence:** Students say a typical day is the equal-share amount, and that sharing \$60 across 4 days means dividing.

**Reasoning:** This evidence connects the definition of mean to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- After 9 games Sofia has scored \_\_\_\_ points. To average 15 across 10 games she needs \_\_\_\_ points in all, so game 10 must be \_\_\_\_ points. Using the balance point instead, her games total \_\_\_\_ points below the target mean and \_\_\_\_ points above it, so game 10 has to land \_\_\_\_ the target mean.

- My answer is \_\_\_\_.

*Mi respuesta es \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.

*Mi afirmación es \_\_\_\_.*

- I know because \_\_\_\_.

*Lo sé porque \_\_\_\_.*

- So \_\_\_\_.

*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
  - **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
  - **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*
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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Mean
2. **Notes 2:** Average
3. **Notes 3:** Measure of center
4. **Notes 4:** Balance point
5. **Notes 5:** Data set
6. **Watch for this mistake:** *The most common mistake with the mean is adding correctly and then dividing by the wrong number — or not dividing at all. Students report the total (for 8, 10, 6, 12 they say the mean is 36) or they divide by a count that does not match the data, such as dividing a 4-value total by 5, or skipping a zero when counting the values. Before you finish, do two things: count the values in the data set out loud, and check that your mean lands between the smallest and largest value. A mean of 36 for data that never goes above 12 is impossible.*
7. **Try It 1:** A. 17 points — *She needs 150 points in all and already has 133, so game 10 must supply  $150 - 133 = 17$  points.*
8. **Try It 2:** A. 24 — *If every value moves up by 4, the balance point moves up by 4 as well, so the new mean is  $20 + 4 = 24$ . In totals: adding 4 to each of  $n$  values adds  $4n$  to the total, and  $4n \div n = 4$ .*